

智慧城市下的智慧窨井监测系统的应用

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摘要: 针对智慧城市建设中各种市政管线窨井井盖丢失、移位、管道溢流、井内有害气体聚集而伴随的安全隐患问题, 探索一种具备信息化管理、智慧化监控、精准化调度的功能的智慧窨井监测系统, 该系统能及时快速处理问题, 提高城市管理效率。智慧窨井监测系统主要包含井盖监测、水位监测、环境温湿度、流量监测、可燃/有害气体监测、水质检测、视频监控等, 根据窨井的性质与需求选择监测类别, 超限发出预警, 及时发送报警信息至平台或相关运维人员的手机 APP。实现“一张图”实时对窨井的全过程管控与危害处理, 做到全面管控, 智慧监管。

关键词: 智慧窨井监测; 智能传感; 蜂窝窄带物联网; 监控云平台

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0 引言

“智慧城市”是由新型网络信息技术支持而发展的一种新型城市管理模式, 终端的感知设备是智慧城市信息资源主要来源。水务、电力、通信、热力、燃气管线循着城市道路分布, 如同人体内血管、神经一样敷设在整个城市, 密度高, 数量大, 各种管网的窨井是管网的重要节点, 智慧窨井监测则是智慧城市的重要部分, 是智慧城市的入口端与信息端。

随着城市规模持续发展, 市政公用设施也在不断完善与扩大, 水务、燃气、电力、通信、热力等部门分管着不同性质的窨井, 窨井与管线分布范围广, 权属复杂, 监管难度大, 传统监管方式浪费大量人力物力, 并且井盖出现丢失、移位、排水管道溢流、井内气体爆炸等现象, 会造成人员伤亡和交通事故, 造成直接或间接的经济损失。各种管线井监管维护, 已成为市政管理的难点与痛点^[1]。

2021年2月9日, 建办督[2021]7号指导意见明确指出: 树立“小井盖、大民生”理念, 始终把群众利益作为推动窨井盖安全管理工作的出发点和落脚点, 到2023年底前, 基本完成各类窨井普查工作, 摸清底数, 健全管理档案, 完成窨井治理专项行动, 窨井安全隐患得到有效治理; 到2025年底前, 窨井安全管理机制进一步完善, 信息化、智能化管理水平明

显加强, 事故风险监测预警能力和应急处置水平显著提升, 窨井安全事故明显减少。

随着智慧城市建设的快速发展, NB-IoT/5G 等网络技术的成熟, 探索开发出一种具备信息化管理、智慧化监控、精准化调度的功能的智慧窨井监测系统, 以满足城市智慧化管理发展需求, 精准监控窨井系统的动态与信息反馈。智慧窨井监测系统主要包含井盖监测、水位监测、环境温湿度、流量监测、可燃/有毒气体监测、水质检测、视频监控等, 根据窨井的性质与需求选择监测类别, 超限发出预警, 及时发送报警信息至平台或相关运维人员的手机 APP, 方便处理信息再反馈至管理云平台。

1 建设意义

(1) 统筹管理, 建立体系。基于各地城市大脑项目, 构建全方位的城市窨井资源状态感知体系, 可对水务管线、照明、热力、燃气、电力及通信等不同行业领域的窨井进行统筹规划, 对各种管线窨井危害信息统一调度处理, 真正形成全市管线窨井的“一盘棋”监督管理。

(2) 全面管控, 智慧监管。将传感、物联网、云计算技术与窨井系统管理紧密结合, 实现对城市窨井的全面感知(智能化)、状态监测(可视化)、故障预警(可控化)、应急调度(闭环化)。建设窨井资源分布图, 实现“一张图”实时对窨井的全过程管控。

(3) 协调联动, 信息管理可实现与各区、各部门及窨井权属单位信息互联互通, 形成协调联动机制。提升多行业、多部门、多领域的系统联控联防能力, 提升

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图 2 智慧窨井监测网络传输系统

2.3 监控平台与管理软件系统

后台服务器收集了大量的窨井感知层的数据,通过对这些数据的分析与处理可以实现以下功能。

一是实时定位监控,可实时监测城市窨井的状态,通过结合管理系统“一张图”系统,对城市内各种窨井进行统一指挥调度与高效维护管养。

二是通过 GIS 地理信息技术在平台电子地图上显示各种窨井基本信息、实时状态、报警信息等,实现集窨井管理、监控和预警“一张网”的综合性平台。

三是报警系统,可对城市窨井的异常情况进行实时监测报警。当采集数据数值超标时,系统能在海量数据的指标展现界面中,按预警规则,将异常数据通过不同的颜色,动态标注出来。让管理人员做到异常数据一目了然。第一时间对井盖的运行状况和异常情况进行监控处理

四是监控平台能够展示报警地址、时间、故障类型等情况,如井内部发生溢水(污水、自来水、雨水等)、环境温度过高(火灾)、烟雾浓度超限(火灾)、有毒气体超限、水质检测等故障情况。

五是鉴权设置,控制中心具有解防功能,只有经授权的工程人员手持终端设备能对窨井系统进行检修与维护。

3 应用场景模拟

例如在一个街区,分别有污水井、燃气井、电力井和运营商的人井等不同功能的地下井,他们所需的报警功能也分别不同,例如污水井,当井下堵塞时,需要我们的报警系统提前告知,这时,水位检测就会产生报警,当水位达到报警高度,我们就可以在污水漫出井盖前,采取相应的措施进行疏通;且在人员下井之前,我们还可以根据有害气体检测模块得知是否可以下井施工,也可以通过视频了解井下的情况。同样,在街区的燃气总阀发生泄露时,我们的告警系统同样会提示相关部门,在发生危险前采取必要的措施;如果街区内的井盖被损坏,产生角度和光照的变化,本系统也会实时告警,避免人员发生意外。

4 结 语

智慧窨井监测系统是智慧城市的重要组成部分。当前各种在线监测设备技术已比较成熟,设备价格有了很大的降幅,很多城市市政管理部门已经建立起智慧监测系统管控平台,随着 5G 与物联网传输技术的普及与应用,智慧窨井监测系统的建设资金已经大幅降低,智慧窨井监测系统能对故障点进行快速判断,提高处理故障速度与效率,实时预警,节省了大量人力物力,提高了城市综合管理的效率和水平,创新了窨井管理新模式,所以智慧窨井监测系统不失为城市市政管理部门未来行之有效的管理系统。

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between the particle shape and particle crushing is quantitatively studied. And the influence of particle crushing on the strength and deformation of rockfill is discussed. The results show that the Duncan–Chang constitutive model can fit the stress–strain curve of artificially simulated rockfill well. Its model parameters conform to the law of natural rockfill. There is a good linear relationship between the particle crushing rate and particle sphericity, and the variation of particle crushing rate with particle sphericity is basically the same under different confining pressures. With the increase of confining pressure, the particle crushing gradually increases and plays a dominant role, which makes the strength of rockfill reduce and the deformation increase.

Keywords: artificial simulated rockfill; particle crushing particle shape; constitutive model

APPLICATION OF ACHIEVEMENTS

Study on Application of Personnel Positioning System in Construction of Smart Construction Site

..... ZANG Xiaolong (234)

Abstract: With the increasing scale and complexity of modern construction projects, the control of construction sites has become increasingly important, and the construction workers are the main body of construction projects. Therefore, the management efficiency of construction workers directly affects the progress and actual results of construction projects. At present, the on-site management of construction projects mainly relies on the manual inspection and inquiry management by the management personnel. However, the mobility of construction site personnel is high, and the energy of management personnel is limited. Therefore, it is important and practical significant to improve the management efficiency of management personnel and speed up the progress of construction projects through the high-tech means and by establishing efficient and intelligent management systems.

Keywords: personnel positioning system; smart construction site; RFID positioning; Bluetooth positioning

Application of Intelligent Manhole Monitoring System in Smart Cities

..... ZHANG Jianqiang, WEI Xiaofei (239)

Abstract: Aiming at the problems of cover loss, displacement, pipeline overflow, inner harmful gas accumulation and the other safety hazards associated with various municipal pipeline manholes in the construction of smart cities, a smart manhole monitoring system with the functions of information management, intelligent monitoring and precise scheduling is explored. This system can promptly and quickly handle the problems and improve the urban management efficiency. The smart manhole monitoring system mainly includes the manhole cover monitoring, water level monitoring, environmental temperature and humidity, flow monitoring, combustible/harmful gas monitoring, water quality detection and video monitoring, etc. The monitoring category is selected according to the characters and demands of manhole. The warning is issued when the limit is exceeded, and the alarm information is promptly sent to the platform or the mobile phone APP of the relevant operation and maintenance personnel. The real-time control and harm treatment of the whole process of the manhole with "one map" can be realized so as to achieve the comprehensive control and intelligent supervision.

Keywords: monitoring of intelligent manhole; smart sensing; cellular narrow band Internet of Things; monitoring cloud platform

Experimental Study and Application of Recycled Aggregate in Road Base Material

..... ZHANG Li (242)

Abstract: The recycled water stabilized base material is prepared by using waste water stabilized recycled aggregate from highway milling instead of natural aggregate, and the engineering application is carried out. The results show that for the waste water stabilized recycled aggregate from the same road, its drying or non-drying treatment has little effect on the optimum moisture content of recycled water stabilized mixture. The optimum moisture content is generally about 4%~5% higher than the water absorption of recycled graded aggregate. For the waste water stabilized recycled aggregate from different road, the higher its crushing value, the lower the unconfined compressive strength, splitting tensile strength and compressive rebound modulus of recycled water stabilized aggregate. The performance of recycled water stabilization is comprehensively considered. The recycled fine aggregate from Road 1 and the recycled